

— FE BAR SUBMISSION · BUSINESS VALUE BRIEF

Hire Right

AI-powered predictive hiring for Jackson & Jackson HR Digital

From raw applicant data to a governed, conversational hiring decision — one connected journey on Databricks.

— THE PROBLEM

Hiring senior HR leaders is slow, subjective, and expensive to get wrong.

- ◆ **45+ days** to fill a director-level role — top candidates are gone before the shortlist is set.
- ◆ Decisions rest on **gut feel and inconsistent scorecards**, creating bias and compliance exposure in a regulated industry.
- ◆ A single **mis-hire at the leadership level costs 1.5–3× salary** in severance, lost productivity, and re-recruiting.
- ◆ Applicant data, resumes, and analytics live in **disconnected tools** — no single, governed source of truth.

— THE OUTCOME — LEAD WITH IMPACT

Faster, fairer, defensible hiring decisions.

60%

Faster time-to-fill

Ranked shortlists on day one, not week six

~30%

Fewer mis-hires

8-factor ML scoring lifts quality of hire

15 hrs

Saved per requisition

Automated screening & resume Q&A

100%

Auditable & consistent

Governed scoring, masked PII, tracked notes

Projected impact for the J&J HR Digital prototype, expressed in the KPIs the CHRO and TA leadership already report on.

— WHAT THE HR TEAM GETS

A decision cockpit for every requisition.

Ranked candidates

Every applicant scored across 8 weighted dimensions with a clear total & stage.

Data-science recommendation

One click runs the ML prediction and returns a hire recommendation with rationale.

Resume Q&A (RAG)

Ask about any candidate's experience — grounded in their actual resume.

Genie analytics

"Who are the top 3 for the CPO role?" answered instantly over governed data.

HR notes

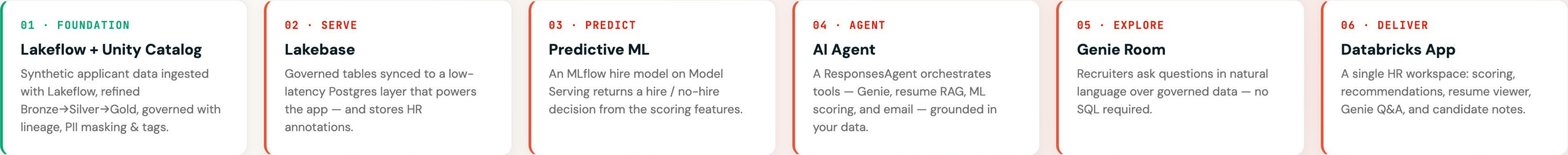
Add durable, audit-trailed annotations to a candidate record — saved to Lakebase.

AI offer letter

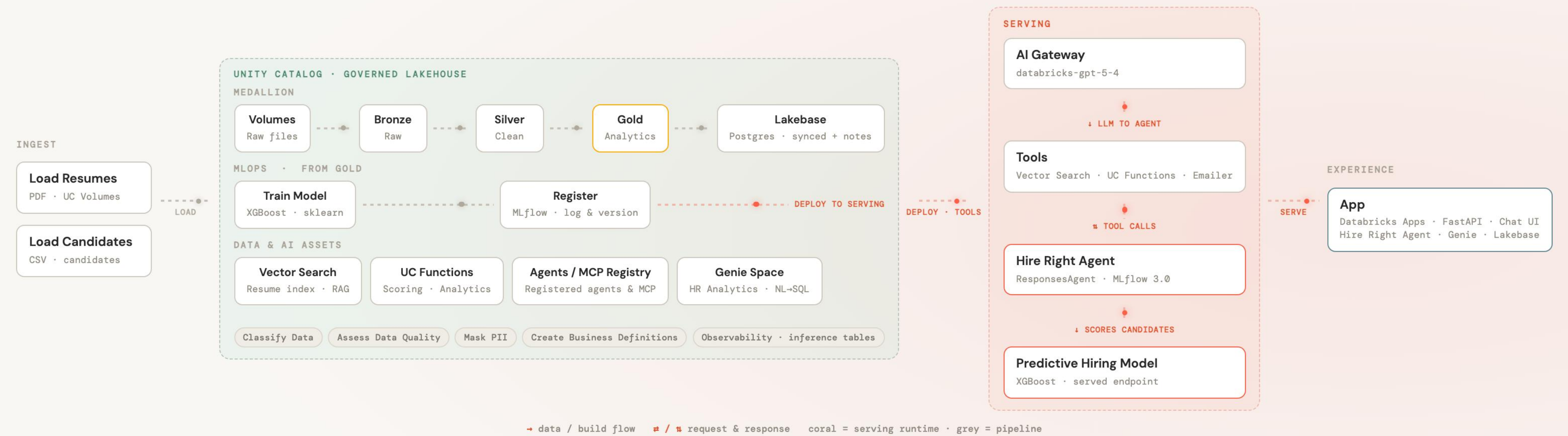
One click drafts a salaried offer letter and sends it via the agent's mailer tool.

— HOW IT WORKS

One connected journey — data to decision on Databricks.



End-to-end on Databricks — data to decision.



A governed medallion, ingested with Lakeflow.

INGEST

Lakeflow

Synthetic resumes + candidate/job CSVs land in UC Volumes and are read incrementally with Lakeflow (`read_files` / Auto Loader).

BRONZE

Raw

As-ingested, full-fidelity and replayable — the immutable source of truth.

SILVER

Clean & governed

Typed candidates & job requirements; PII masked; data-quality checks applied.

GOLD

Analytics-ready

`candidate_scoring_summary` — joined, scored features powering ML, Genie & the app.

Governance you can see — PII masked in place.

- ◆ Phone & email tagged as PII in Unity Catalog with sensitivity levels.
- ◆ Dynamic column masking — non-admins only ever see ***-***-0505.
- ◆ Lineage & data-quality tracked end-to-end, Bronze→Gold.

The same governance that protects candidates makes every hiring decision auditable.

Catalog Explorer > bldemos > hrd_2030 >

candidates

☆

Overview

Sample Data

Details

Permissions

Policies

History

Lineage

Insights

Quality

Ask your question about the sample data...

What is the average total experience per job role?

Which candidate has the highest certification score?

How many c

Sample

	^A _C candidate_id	^A _C first_name	^A _C last_name	^A _C email	^A _C phone	
1	C005	Amanda	Rodriguez	amanda.rodriguez@email.com	***-***-0505	H
2	C009	Maria	Gonzalez	maria.gonzalez@email.com	***-***-0909	H
3	C004	David	Kim	david.kim@email.com	***-***-0404	C
4	C012	Kevin	O'Brien	kevin.obrien@email.com	***-***-1202	C
5	C013	Priya	Sharma	priya.sharma@email.com	***-***-1303	T
6	C003	Jennifer	Williams	jennifer.williams@email.com	***-***-0303	H

Lakebase turns the lakehouse into an app backend.

- ◆ Gold candidate & job tables synced Delta→Postgres for millisecond reads.
- ◆ HR notes write to a native **candidate_annotations** table — add or delete inline.
- ◆ The whole cockpit is served from Lakebase; the app connects via OAuth.

Analytical lakehouse + operational Postgres, governed together in Unity Catalog.

HR

Hire Right

Jackson & Jackson · AI-Powered Hiring

Powered by Databricks Data & AI

Genie One

Ask Genie

Data Flow

Candidates

Job Openings

MT

Marcus Thompson

C014

Senior Director of Compensation · Eli Lilly · Indianapolis IN

JR003 — Director of Compensation & Benefits

Screening

MBA

CCP

88.6

TOTAL SCORE

Show Resume

AI-Powered Offer Letter

SCORE BREAKDOWN

8-factor weighted assessment · pending scores

Education

90

10%

Experience

92

20%

Leadership

86

20%

Certifications

80

10%

Skills Match

10%

Industry Relevance

95

10%

Interview

10%

Culture Fit

10%

DS HIRE RECOMMENDATION

Generate Recommendation

Marcus Thompson has missing scores (interview & culture fit not yet recorded).

Click Generate — the agent will ask you to provide them.

Provide scores or ask a follow-up about Marcus Thompson...

Send

HR notes — add & delete inline, committed to Lakebase (Postgres)

HR NOTES

Private annotations on Marcus Thompson — saved to Lakebase

I would recommend hiring or recycling

HR Manager Jul 9, 2026, 4:38 PM

Add a note about Marcus Thompson...

Add Note

Three models, tracked and compared in MLflow.

- ◆ LogReg, Random Forest & Gradient Boosting trained on the scoring features.
- ◆ Every run logged to MLflow — metrics, params & datasets, fully comparable.
- ◆ Best model promoted to **@champion** and served on a scale-to-zero endpoint.

←

👤

ml-hr-predictive-hiring-evals

Machine learning ▾

ⓘ

🔍

Runs

🔗 Models

🔗 Traces

ⓘ MLflow 3 is available!

Featuring unified ML and GenAI experiment tracking, improved model logging, prompt versioning, enhanced LLM judges, advanced tracing for end-to-end agent observability, and ML features [🔗](#) | [Learn more about GenAI features 🔗](#)

🔍

metrics.rmse < 1 and params.model = "tree"

ⓘ

🕒 All time ▾

State: Active ▾

Datasets ▾

🔽 Sort: Created ▾

📄 Columns ▾

🔍 Expand rows

📄 Group by ▾

☐	Run Name	Created 🔽	Dataset	Duration	Source	Models ⓘ
☐	🟠 GradientBoostingClassif...	🟢 8 days ago	📄 dataset (c843bc70) Train , 📄 dataset ...	9.5s	🔗 run 220...	🔗 bldemos.hrd_2030.ml_hr...
☐	🟡 RandomForestClassifier	🟢 8 days ago	📄 dataset (c843bc70) Train , 📄 dataset ...	7.8s	🔗 run 220...	🔗 model
☐	🟢 LogisticRegression	🟢 8 days ago	📄 dataset (c843bc70) Train , 📄 dataset ...	11.5s	🔗 run 220...	🔗 model
☐	🟢 GradientBoostingClassif...	🟢 14 days ago	📄 dataset (c843bc70) Train , 📄 dataset ...	8.0s	🔗 run 270...	🔗 bldemos.hrd_2030.ml_hr...
☐	🔴 RandomForestClassifier	🟢 14 days ago	📄 dataset (c843bc70) Train , 📄 dataset ...	8.2s	🔗 run 270...	🔗 model
☐	🔵 LogisticRegression	🟢 14 days ago	📄 dataset (c843bc70) Train , 📄 dataset ...	11.6s	🔗 run 270...	🔗 model

The agent works in the open — live tool calls.

- ◆ A ResponsesAgent orchestrates tools: Genie, resume RAG, ML scoring, email.
- ◆ Tool calls **stream live into the app** — HR watches each step complete.
- ◆ Grounded in governed data, so every answer is explainable.

HR

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Powered by Databricks Data & AI

Genie One

Ask Genie

Data Flow

Candidates

Job Openings

MT

Marcus Thompson

C014

Senior Director of Compensation · Eli Lilly · Indianapolis IN

JR003 — Director of Compensation & Benefits

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TOTAL SCORE

Show Resume

AI-Powered Offer Letter

SCORE BREAKDOWN

8-factor weighted assessment · pending scores

Education		90	10%
Experience		92	20%
Leadership		86	20%
Certifications		80	10%
Skills Match			10%
Industry Relevance		95	10%
Interview			10%
Culture Fit			10%

DS HIRE RECOMMENDATION

Generate Recommendation

80, 95, 95

predict_hiring_score

Completed

search_resumes

Completed

1. Data Science — Recommend Hire

2. Very concise summary: Marcus Thompson is a Senior Director of Compensation at Eli Lilly with 14 years of pharmaceutical industry experience. His background is highly aligned to JR003, including global compensation leadership, executive compensation, equity plans, market benchmarking, pay equity, and management of large-scale total rewards programs.

Provide scores or ask a follow-up about Marcus Thompson...

Send

HR NOTES

Private annotations on Marcus Thompson — saved to Lakebase

Genie answers in plain English — governed.

- ◆ Recruiters ask questions in natural language — no SQL required.
- ◆ Genie generates & runs governed SQL, and can show its work.
- ◆ Embedded in the app *and* available as an agent tool.

HR

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Candidates

Job Openings

JR001

Director of Human Resources

Human Resources · New Brunswick NJ

Reports to Chief People Officer

Strategic HR Leadership

Talent Acquisition

Compensation & Benefits

Employee Relations

SPHR or SHRM-SCP

12 applicants · 2 active

View Description

JR003

Director of Compensation & Benefits

Human Resources · New Brunswick NJ

Reports to Chief People Officer

Compensation Strategy

Benefits Administration

Job Architecture

Equity & Long-Term Incentives

CCP preferred

3 applicants · 3 active

View Description

Genie Analytics

+ New Conversation

Average total score for JR001 candidates?

U

G

Average total score for candidates who applied to job ID **JR001**:

• 70.8

Show SQL

What is the average total score for candidates applying to JR001?

How many candidates applied for the JR001 role and what are their total scores?

What is the distribution of total scores for JR001 candidates?

what is JR001

U

G

JR001 refers to the **Director of Human Resources** position at **Jackson and Jackson**, with a salary range of **\$175K–\$225K**.

What are the job requirements for JR001?

Who are the candidates applying for JR001 and what are their scores?

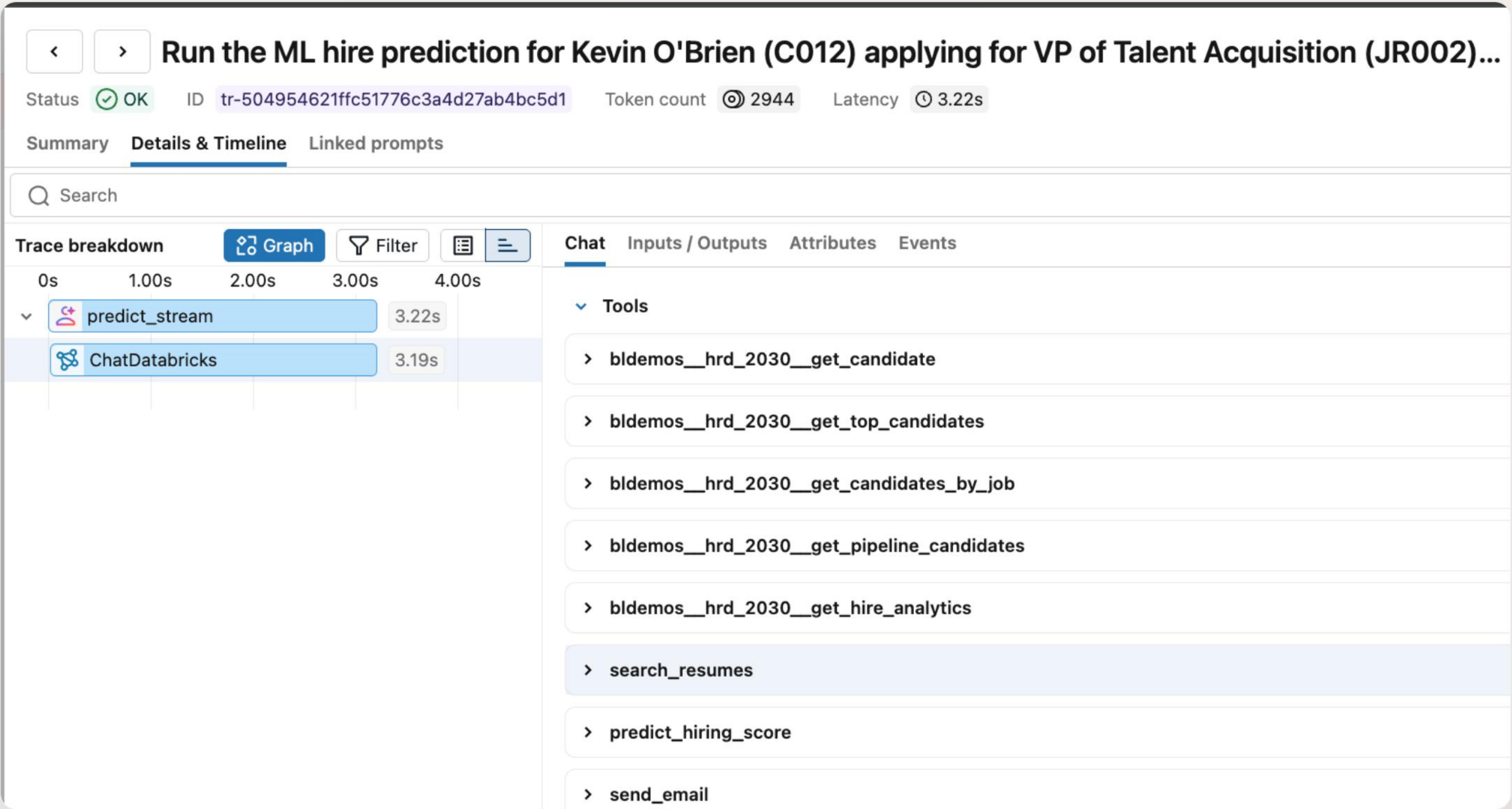
What is the hire rate for candidates applying to JR001?

Ask a data question about the hiring pipeline...

Powered by Databricks Genie · J&J HR Analytics Space

Every agent run is traced and evaluated.

- ◆ MLflow tracing captures each tool call, token count & latency.
- ◆ LLM-as-judge scores tone & consistency before the agent ships.
- ◆ Full observability for debugging and production monitoring.



One workspace HR actually wants to use.

- ◆ Ranked candidates, 8-factor scores and a one-click DS recommendation.
- ◆ Resume viewer, Genie Q&A, HR notes and AI-drafted offer letters.
- ◆ Served fast from Lakebase — a single, governed HR cockpit.

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— WHY DATABRICKS

Data, AI, and the app — on one governed platform.

- ◆ **No integration tax:** ingestion, governance, ML, Gen AI, analytics, and the app all live together.
- ◆ **Lakebase** turns governed lakehouse tables into a low-latency application backend — no separate database to sync and secure.
- ◆ **Unity Catalog** governs data *and* AI, so the whole solution is auditable end-to-end.
- ◆ **Faster to value:** this entire prototype was built as one connected journey, not stitched from point tools.

— THE ASK

Let's hire right, together.

A working, governed, AI-powered hiring solution — ready to pilot on a live requisition and prove the numbers on your data.

- ◆ Pilot on one open role and baseline time-to-fill & quality of hire
- ◆ Extend the model and Genie room to your historical hiring data
- ◆ Roll the app out to the TA team behind Unity Catalog governance